

OSAMA ALI NAJI

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PROFILE

AI Engineer with First-Class Honors in Computer Science and hands-on experience building and deploying production AI systems. Specialize in ML & DL, Agentic AI, RAG pipelines, LLM integration, and Computer Vision — delivering end-to-end solutions using LangChain, n8n, PyTorch, and YOLO. Proven ability to work independently, having designed ML & DL models, AI automation agents, and deep learning systems at Tytan KSA. Actively building multi-agent systems and real-world AI products. Dedicated to deploying scalable AI solutions that drive operational efficiency and business growth.

EDUCATION

Taibah University

Jan 2020 – May 2024

Bachelor of Science in Computer Science

- GPA: **4.95/5.00** - First Class Honors.
- Graduation Project: Phishing Attack Detection Using AI.
- Coursework: Data Structures & Algorithms, Operating Systems, Networks, Software Engineering.

KAUST (King Abdullah University of Science and Technology)

Oct 2024 – Feb 2025

Specialization — Artificial Intelligence

- Focused study in AI concepts, tools, and practical applications.

Apple Developer Academy | TUWAIQ

Oct 2025 – Nov 2025

Education Scholarship — iOS Dev, Swift, UI/UX

- Scholarship program in app development, Swift coding, and UI/UX design fundamentals.

WORK EXPERIENCE

AI Engineer

Aug 2025 – Present

Tytan KSA | Full-time | On-site

- Independently led the AI division — designed, built, and deployed ML/DL models for delivery performance optimization with no team or senior guidance.
- Built ML models for delivery time prediction and route optimization, improving operational decision-making.
- Developed Power BI dashboards that gave management real-time visibility into delivery KPIs and bottlenecks.
- Built AI automation agents using n8n to streamline internal workflows and reduce manual operations.

AI, Backend & Database Intern

Mar 2025 – Jun 2025

SIT – IT Solutions | Internship

- Gained hands-on experience in database engineering, data science, AI integration, and backend development.
- Built AI-integrated applications, designed relational databases, and developed RESTful APIs.

Research & Development Leader

Nov 2024 – Jun 2025

Wae (وعي) | Volunteer | Riyadh, Hybrid

- Led AI awareness workshops targeting diverse community audiences across Saudi Arabia.
- Directed AI-driven R&D initiatives and developed educational content on emerging AI technologies.

Artificial Intelligence Intern

Jul 2023 – Aug 2023

Taibah Valley | Internship | Medina

- Developed a high-accuracy Arabic sentiment analysis model using Python, NLTK, and scikit-learn.
- Conducted data acquisition, cleaning, and preprocessing pipelines for NLP model training.

PROJECTS

Hotel Customer Service Agent — WhatsApp AI Bot | *n8n · LangChain · RAG · WhatsApp API · OpenAI API · Chroma DB*

- **Built an end-to-end Agentic AI system** that handles hotel customer inquiries automatically via WhatsApp — room availability, pricing, booking FAQs, and escalation to human staff.
- **Implemented RAG pipeline** connecting hotel knowledge base (policies, room types, services) to an LLM, enabling accurate and context-aware responses in Arabic and English.
- **Orchestrated the full workflow** using n8n — integrating WhatsApp Business API, vector database retrieval, LLM response generation, and fallback logic in a single automated pipeline.

Underwater Scene Segmentation — Multi-Class Computer Vision | *PyTorch · U-Net · EfficientNet-B1 · SUIM Dataset · Semantic Segmentation*

- **Fine-tuned a pretrained U-Net model** with EfficientNet-B1 encoder on the SUIM dataset for 8-class underwater semantic segmentation (divers, fish, reefs, robots, seafloor, and more).
- **Built full ML pipeline:** custom PyTorch Dataset class, image/mask transforms, train/validation loops with CrossEntropyLoss and AdamW optimizer.
- **Implemented prediction visualization** and loss curve analysis — demonstrates strong Computer Vision and deep learning engineering skills.

YouHealth — AI Medical Chatbot 2nd Place, SoundAI Hackathon (SDAIA) | *Python · LLM · lablab MENA / SDAIA*

- Built AI assistant that converts patient conversations into structured medical reports.
- Nominated for GAIA AI Accelerator. Collaborated on full system design, NLP pipeline, and deployment.

AI-Powered Crowd & Traffic Monitoring System | *YOLOv8 · MoViNet · SORT · Python · Computer Vision*

- Real-time smart city AI system combining object detection, action recognition, and zone-based heatmaps.
- Designed for safety monitoring and traffic analytics — demonstrates multi-model Computer Vision pipeline.

PhishWarden — Phishing Detection System (Graduation Project) | *Python · Flask · ML · NLP · HTML/CSS/JS*

- ML-based system detecting phishing emails and URLs via NLP feature extraction and classification.
- Full-stack implementation with Flask backend — key contributor to (A+) and GPA 4.95/5.0 First-Class Honors.

SKILLS

ML / DL Engineering	TensorFlow · scikit-learn · NLP · NLTK · Hugging Face · Deep Learning · Python · Git SQL · Flask · FastAPI · REST APIs · Linux · Power BI · Cloud Deployment AWS · Docker
Computer Vision	YOLO · U-Net · Semantic Segmentation · Object Detection · PyTorch
Agentic AI	n8n (workflows & custom nodes) · LangChain Agents · CrewAI · AutoGen · OpenWebUI
LLMs & RAG	RAG Pipelines · Prompt Engineering · OpenAI API · Ollama · ChromaDB · Embeddings
Soft Skills	Self-directed · Problem Solving · Leadership · Communication · Adaptability

CERTIFICATES & TRAINING COURSES

• Apple Developer Academy — Apple	Nov - 2025
• Machine Learning Specialization — Stanford	Dec - 2025
• IBM Machine Learning Professional Certificate — IBM	May - 2025
• Associate AI Engineer for Data Scientists — DataCamp	Jul - 2025
• Deep Learning Specialization — DeepLearning.AI	May - 2025
• AI Program (Introductory & Advanced) — KAUST	Feb - 2025
• Python 3 Programming — University of Michigan	Nov - 2024
• McKinsey Forward Program — McKinsey Company	Dec - 2024
• Track of Artificial Intelligence — FutureX	Mar - 2023